

Independent Tunability of Local Curvature and Global Holonomy in a Source-Free $\mathrm{SO}(2)$ Gauge Field

Anthony Vito Coppa

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Abstract

A gauge connection's field strength and its holonomy are not the same fact told twice. We exhibit an explicit $\mathrm{SO}(2)$ connection on $\mathbb{R}_t \times S_\theta^1 \times \mathbb{R}_x \times \mathbb{R}_y$ with a genuine Lorentzian metric, and prove two independence results by direct computation, not analogy. First: the Maxwell curvature is fixed entirely by a free function f of retarded time, while the holonomy around the compact fiber is fixed entirely by an unrelated real parameter a — one is a local differential fact, the other a global integral fact, and the construction sets them to arbitrary independent values simultaneously. Second: on the same background, two worldlines with the same endpoints in coordinate time carry different proper times while carrying independently specifiable holonomy — curvature-free local geometry and global phase transport are not the same measurement wearing two names. There is no interpretation attached to either result beyond what the equations say.

1 Setup

Let $V = \mathbb{R}^2$ carry the standard Euclidean inner product, and let

$$C_1 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \in \mathfrak{so}(2) \quad (1)$$

generate finite rotations $U(\theta) = e^{\theta C_1} = I \cos \theta + C_1 \sin \theta$.

Let $M = \mathbb{R}_t \times S_\theta^1 \times \mathbb{R}_x \times \mathbb{R}_y$. Let η be the closed 1-form on S^1 with $\oint_{S^1} \eta = 2\pi$. For constants $c, \rho > 0$, equip M with the Lorentzian metric

$$g = -c^2 dt^2 + \rho^2 \eta^2 + dx^2 + dy^2. \quad (2)$$

Let $E = M \times V$ be an oriented rank-2 real vector bundle with structure group $\mathrm{SO}(2)$. For $f \in C^2(\mathbb{R})$ with $f' \not\equiv 0$ and retarded coordinate $u = t - x/c$, define the one-parameter connection family

$$\mathcal{A}_a = a \eta + f(u) dy, \quad a \in \mathbb{R}. \quad (3)$$

2 Curvature and Holonomy are Independent

Proposition 2.1 (Curvature). The field strength of $\mathcal{A}_a$ is

$$F = d\mathcal{A}_a = f'(u) du \wedge dy, \quad (4)$$

with components $F_{ty} = f'(u)$, $F_{xy} = -\frac{1}{c} f'(u)$. It satisfies the source-free Maxwell equations exactly:

$$\nabla_\mu F^{\mu y} = \partial_t F^{ty} + \partial_x F^{xy} = -\frac{1}{c^2} f''(u) + \frac{1}{c^2} f''(u) = 0, \quad dF = 0, \quad d*_g F = 0. \quad (5)$$

F depends on $\mathcal{A}_a$ only through f ; it does not depend on a .

Proposition 2.2 (Holonomy). For a spatial loop l winding once around S_θ^1 at fixed t, x, y ,

$$\text{Hol}_{\mathcal{A}_a}(l) = \exp\left(-C_1 \oint_l \mathcal{A}_a\right) = e^{-2\pi a C_1}, \quad \Delta_{\text{phase}} := \text{Hol}_{\mathcal{A}_a}(l) - I = e^{-2\pi a C_1} - I. \quad (6)$$

This depends on $\mathcal{A}_a$ only through a ; it does not depend on f .

Theorem 2.3 (Independent Tunability). F and Δ_{phase} are functions of disjoint, freely chosen data: any $f \in C^2(\mathbb{R})$ with $f' \not\equiv 0$ may be paired with any $a \in \mathbb{R}$, giving an arbitrary curvature profile and an arbitrary holonomy value on the same source-free connection, simultaneously and without constraint on one another.

3 Proper Time and Gauge Phase are Independent

For endpoints $p = (0, [0], 0, 0)$ and $q = (T, [0], 0, 0)$ with $T > \frac{2\pi\rho}{c}$, define timelike paths

$$\gamma_0(t) = (t, [0], 0, 0), \quad \gamma_1(t) = \left(t, \left[\frac{2\pi t}{T}\right], 0, 0\right). \quad (7)$$

Proposition 3.1 (Proper Time).

$$\tau[\gamma_0] = T, \quad \tau[\gamma_1] = T \sqrt{1 - \left(\frac{2\pi\rho}{cT}\right)^2}, \quad \Delta_{\text{time}} := \tau[\gamma_0] - \tau[\gamma_1] \neq 0. \quad (8)$$

Theorem 3.2 (Coexistence). $\Delta_{\text{time}} \in \mathbb{R}$ and $\Delta_{\text{phase}} \in \text{End}(E_p)$ coexist on the same pair of shared-endpoint worldlines without functional dependence: Δ_{time} is fixed by ρ and T alone, and Δ_{phase} by a alone, via the same connection family of Section 2.